AP Microeconomics Curve Challenge

Answer Key and Explanations

Solutions

1. **Answer: C**. Because supply is perfectly inelastic at Q_f, a rightward shift in demand raises price but cannot increase quantity. The new equilibrium is at price P_2 and quantity Q_f.
2. **Answer: B**. With fixed seating, quantity supplied remains Q_f. At the ceiling price P_c, quantity demanded is larger than Q_f, so a shortage results even though the number of seats sold stays fixed at Q_f.
3. **Answer: A**. Total revenue is price times quantity. Quantity stays fixed at Q_f, while price rises from P_1 to P_2, so revenue increases.
4. **Answer: D**. A binding ceiling below equilibrium encourages nonprice rationing and can create resale or scalping. Some consumers may ultimately pay above the legal ceiling in secondary transactions.
5. **Answer: E**. At the free-trade world price P_w, domestic quantity supplied is Q_1 and domestic quantity demanded is Q_4. Imports therefore equal $Q_4 - Q_1$.
6. **Answer: B**. At the tariff-inclusive price P_t, domestic supply rises to Q_2 and domestic demand falls to Q_3. Imports shrink to $Q_3 - Q_2$.
7. **Answer: B**. Tariff revenue equals the tariff per unit times the number of imported units after the tariff. That is $(P_t - P_w)(Q_3 - Q_2)$.
8. **Answer: C**. The higher domestic price benefits domestic producers, and the government collects tariff revenue. Domestic consumers are harmed because they pay a higher price and buy less.
9. **Answer: A**. The graph shows the consumer-price increase is larger than the producer-price decrease, so consumers bear the larger share. Tax incidence depends on elasticities, not on the legal assignment of the tax.
10. **Answer: C**. Tax revenue is the per-unit tax wedge multiplied by the quantity sold after the tax. The vertical wedge is $P_c - P_p$, and the post-tax quantity is Q_t.
11. **Answer: A**. The tax eliminates mutually beneficial trades from Q_t to Q^*. The deadweight loss is the triangle with height equal to the tax wedge and base equal to the lost quantity.
12. **Answer: A**. When supply is more elastic, producers are more able to reduce quantity supplied in response to the tax. The less elastic side of the market then bears more of the burden, so consumers pay a larger share.
13. **Answer: A**. A competitive firm chooses the quantity where price equals marginal cost on the upward-sloping portion of MC. At P_1, that output is Q_1, and the price line lies above ATC there, so economic profit is positive.
14. **Answer: B**. The shutdown rule compares price with AVC, not ATC. Since P_2 lies above AVC and intersects MC at Q_2, the firm continues producing Q_2 in the short run while taking a loss.
15. **Answer: C**. A change in fixed cost shifts ATC but does not shift MC or AVC. Because the firm's output rule is $P = MC$ as long as price covers AVC, the output remains Q_2.
16. **Answer: B**. Positive economic profit attracts entry. Entry shifts market supply right and drives the market price downward until firms earn zero economic profit in the long run.
17. **Answer: B**. A single-price monopolist produces where $MR = MC$, which is Q_m, and then charges the price consumers are willing to pay on the demand curve, which is P_m.
18. **Answer: B**. Allocative efficiency requires price to equal marginal cost. At the monopoly quantity Q_m, the price on the demand curve is above MC, so the monopoly underproduces relative to the efficient level.
19. **Answer: C**. A perfectly price-discriminating monopolist sells every unit for which willingness to pay is at least marginal cost. That expands output to Q_c, where demand intersects MC, the same quantity as in perfect competition.
20. **Answer: C**. Average-total-cost pricing increases output relative to an unregulated monopoly because price must fall. But since the regulated price still exceeds MC at the chosen quantity, output remains below the allocatively efficient level Q_c.

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21. **Answer: B**. In long-run monopolistic competition, entry drives economic profit to zero. Graphically, that appears as the demand curve tangent to ATC at the firm's chosen quantity.

22. **Answer: B**. Excess capacity means the firm produces less than the output that minimizes average total cost. The graph shows the firm's output Q_m is to the left of Q_e, the minimum-ATC quantity.

23. **Answer: C**. Allocative efficiency requires $P = MC$. In monopolistic competition, the downward-sloping demand curve implies price exceeds marginal cost at the profit-maximizing output.

24. **Answer: A**. Positive short-run profits attract entry of close substitutes. That reduces demand for this firm's product and shifts its demand curve left until economic profit is eliminated.

25. **Answer: B**. A rightward shift in labor demand raises the equilibrium wage and the equilibrium quantity of labor employed in a competitive market.

26. **Answer: B**. With a binding minimum wage, firms hire only the quantity of labor demanded, which is Q_d. Workers supply Q_s, so unemployment equals $Q_s - Q_d$.

27. **Answer: C**. Total earnings for employed workers equal the wage times the number of workers actually employed. Under the binding minimum wage, that is W_m times Q_d.

28. **Answer: B**. If labor demand is relatively inelastic, the wage increase outweighs the reduction in employment, making total earnings more likely to rise.

29. **Answer: B**. A monopsonist hires labor where $MRP = MFC$, which determines L_m. It then pays the wage found on the labor supply curve at that employment level, which is W_m.

30. **Answer: C**. In the competitive benchmark, wage and employment are W_c and L_c. The monopsonist instead chooses W_m and L_m, both of which are lower.

31. **Answer: C**. A minimum wage set at the competitive wage can flatten the firm's marginal factor cost up to the relevant range. The monopsonist then hires labor where that wage equals MRP, increasing employment to L_c.

32. **Answer: C**. In a monopsony, to attract another worker the employer generally must offer a higher wage, and that higher wage applies to all previously hired workers. Therefore the marginal cost of an extra worker exceeds the wage itself.

33. **Answer: C**. Private firms ignore external costs and produce where demand intersects MPC, which is the market quantity Q_m. The socially efficient quantity is lower, at Q_s.

34. **Answer: B**. The corrective tax must internalize the marginal external cost at the socially efficient output. Graphically, that is the vertical gap between MSC and MPC at Q_s.

35. **Answer: C**. Overproduction from Q_s to Q_m creates a triangle of lost welfare. On those excess units, marginal social cost exceeds marginal social benefit.

36. **Answer: B**. The tax reduces overproduction and moves the market to the socially efficient quantity Q_s. Eliminating the deadweight loss increases total surplus.

37. **Answer: C**. A Lorenz curve closer to the 45-degree line indicates a more equal distribution of income. Country B's curve is consistently closer to that line than Country A's.

38. **Answer: B**. The Gini coefficient rises as the Lorenz curve bows farther away from the line of equality. Since Country A is more bowed out, Country B has the smaller Gini coefficient.

39. **Answer: B**. Redistribution that makes lower-income households receive a larger share of total income moves the Lorenz curve closer to the line of equality.

40. **Answer: D**. If the bottom 80% receive 50%, the remaining top 20% must receive the other 50% of total income.

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